**17** $x^2 - 9y^2 = 0$ where $x > 0$ and $y > 0$

(a) Work out the ratio $x : y$

$x^2 - 9y^2 = (x - 3y)(x + 3y) = 0.$

**1 mark**
Difference of squares

$x > 0$ and $y > 0$, so $x + 3y > 0$.

Therefore $x - 3y = 0$, giving $x = 3y$.

**1 mark**
Positive root x=3y

$x : y = 3 : 1.$

**1 mark**
Ratio3:1

**(3)**

(b) Simplify fully $\dfrac{3 - 4x - 4x^2}{2x^2 - 7x + 3}$

Numerator

| × | $2x$ | $3$ |
|---|---|---|
| $-2x$ | $-4x^2$ | $-6x$ |
| $1$ | $2x$ | $3$ |

Denominator

| × | $2x$ | $-1$ |
|---|---|---|
| $x$ | $2x^2$ | $-x$ |
| $-3$ | $-6x$ | $3$ |

**1 mark**
(3+2x)(1−2x)

**1 mark**
(2x−1)(x−3)

$$\frac{-(3 + 2x)(2x - 1)}{(2x - 1)(x - 3)} = \frac{3 + 2x}{3 - x}$$

**1 mark**
Signs and cancellation

Cancel2x−1; $x \neq \frac{1}{2}$ and $x \neq 3$.

**(3)**

**(Total for Question 17 is 6 marks)**